

HF Controls Ltd

100 Albert Drive, Glasgow, G41 2SJ

www.hf-group.co.uk

Call 0800 783 9480

Glasgow

Edinburgh

Manchester

Belfast

Aberdeen

PROJECT: RSGT, LIVERPOOL

PROJECT LOCATION: SWITCHROOM 2

PROJECT REFERENCE NUMBER: 284595

CLIENT: PEEL PORTS

CLIENT ADDRESS: MARITIME CENTRE PORT OF LIVERPOOL

CLIENT ORDER NUMBER: ????

CONTROL SYSTEM DETAILS & SPECIFICATION

SYSTEM TYPE:	SAFE AREA
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ENCLOSURE MATERIAL:	MILD STEEL
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RATED VOLTAGE: 230V AC

RATED FREQUENCY:	50 Hz
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RATED AMPERE:	6A
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CONTROL VOLTAGE: 24V DC

DEGREE OF PROTECTION:	IP 55
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PAINT FINISH: POWDER COATED RAL7035

YEAR OF MANUFACTURE:	2022
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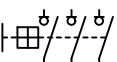
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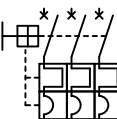
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00	12/01/24	AS FITTED		INSTALL	PEEL PORTS	DRAWN	D.BALDOCK	
01	23/09/24	PRE-START SOUNDER / BEACON ADDED				CHECKED	M.WRIGHT	
02	09/01/25	BELT SOUNDERS ADDED. FIRE DOOR SWITCH FROM MCL ADDED ISOLATOR AUX STATUS AND S/STOPS ADDED				DATE	11/05/2023	
03			TITLE	RSGT LIVERPOOL SWITCHROOM 2	DRG. No.:			
04					284595_4101			
05								
06			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :		PAGE No.:	
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A	B	C	D	E	F																																																
<table><tr><th>CURRENT CIRCUIT</th><th>CONDUCTOR</th><th>WIRING COLOUR</th><th>MINIMUM CROSS-SECTION</th></tr><tr><td>3-PHASE POWER</td><td>L1</td><td>BROWN</td><td>2.5mm2</td></tr><tr><td>3-PHASE POWER</td><td>L2</td><td>BLACK</td><td>2.5mm2</td></tr><tr><td>3-PHASE POWER</td><td>L3</td><td>GREY</td><td>2.5mm2</td></tr><tr><td>3-PHASE NEUTRAL</td><td>N</td><td>LIGHT BLUE</td><td>2.5mm2</td></tr><tr><td>1-PHASE POWER</td><td>L1</td><td>BROWN</td><td>1.5mm2</td></tr><tr><td>1-PHASE POWER</td><td>N</td><td>LIGHT BLUE</td><td>1.5mm2</td></tr><tr><td>24VDC</td><td>+</td><td>DARK BLUE</td><td>0.75mm2</td></tr><tr><td>24VDC</td><td>-</td><td>DARK BLUE</td><td>0.75mm2</td></tr><tr><td>MEASURING</td><td>+/-</td><td>GREY</td><td>0.5mm2</td></tr><tr><td>EXTERNAL VOLTAGE</td><td></td><td>ORANGE</td><td>1.5mm2</td></tr><tr><td>PROTECTIVE EARTH CONDUCTOR</td><td>PE</td><td>GREEN/YELLOW</td><td>1.5mm2</td></tr></table>						CURRENT CIRCUIT	CONDUCTOR	WIRING COLOUR	MINIMUM CROSS-SECTION	3-PHASE POWER	L1	BROWN	2.5mm2	3-PHASE POWER	L2	BLACK	2.5mm2	3-PHASE POWER	L3	GREY	2.5mm2	3-PHASE NEUTRAL	N	LIGHT BLUE	2.5mm2	1-PHASE POWER	L1	BROWN	1.5mm2	1-PHASE POWER	N	LIGHT BLUE	1.5mm2	24VDC	+	DARK BLUE	0.75mm2	24VDC	-	DARK BLUE	0.75mm2	MEASURING	+/-	GREY	0.5mm2	EXTERNAL VOLTAGE		ORANGE	1.5mm2	PROTECTIVE EARTH CONDUCTOR	PE	GREEN/YELLOW	1.5mm2
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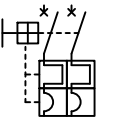
CODE OF ELECTRICAL SYMBOLS



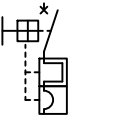
Q = MAIN ISOLATING SWITCH



QFM = ISOLATING CIRCUIT BREAKER
(3 POLE)



QFM = ISOLATING CIRCUIT BREAKER
(2 POLE)



QFM = ISOLATING CIRCUIT BREAKER
(1 POLE)



CONTACT (POWER)
NORMALLY OPENED



CONTACT-THERMOSTAT
NORMALLY CLOSED



FU = OVERLOAD PROTECTIVE DEVICE
(3 POLE)



SF = SELECTOR SWITCH
2 POSITION / STAY PUT



SF = PUSH BUTTON / SPRING RETURN
NORMALLY OPEN



SF = PUSH BUTTON / SPRING RETURN
NORMALLY CLOSED



Kx = CONTROL COIL



PLC INPUT



PLC OUTPUT



INDICATION LED



HEATER

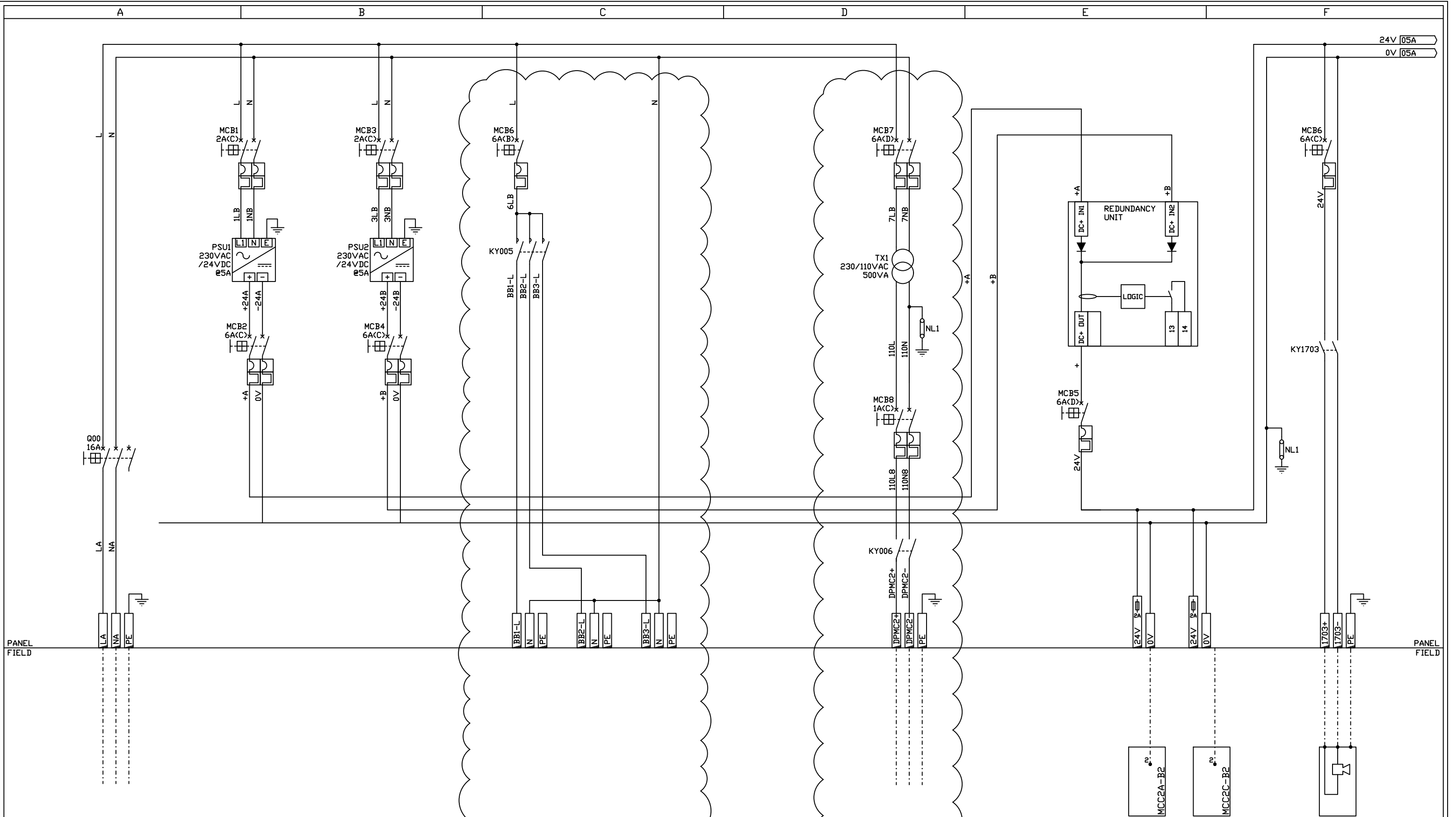


RESISTOR



PSU = AC TO DC POWER CONVERSION

REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BALDOCK	
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03			INSTALL	RSGT LIVERPOOL SWITCHROOM 2	DRG. No.: 284595_4101		
04							
05							
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MAINS INCOMING SUPPLY 1PH/N/E 230VAC 32A		24VDC POWER SUPPLY 1		24VDC POWER SUPPLY 2		BB1 A/B/C BELT SOUNDERS		BB2 A/B/C BELT SOUNDERS		BB3 A/B/C BELT SOUNDERS		MC2 DCE DUST PLANT 110VAC SUPPLY		24VDC REDUNDANCY UNIT		24VDC TO MCC2A		24VDC TO MCC2C		BB1/BB2/BB3 PRE-START SOUNDER SUPPLY 24VDC	
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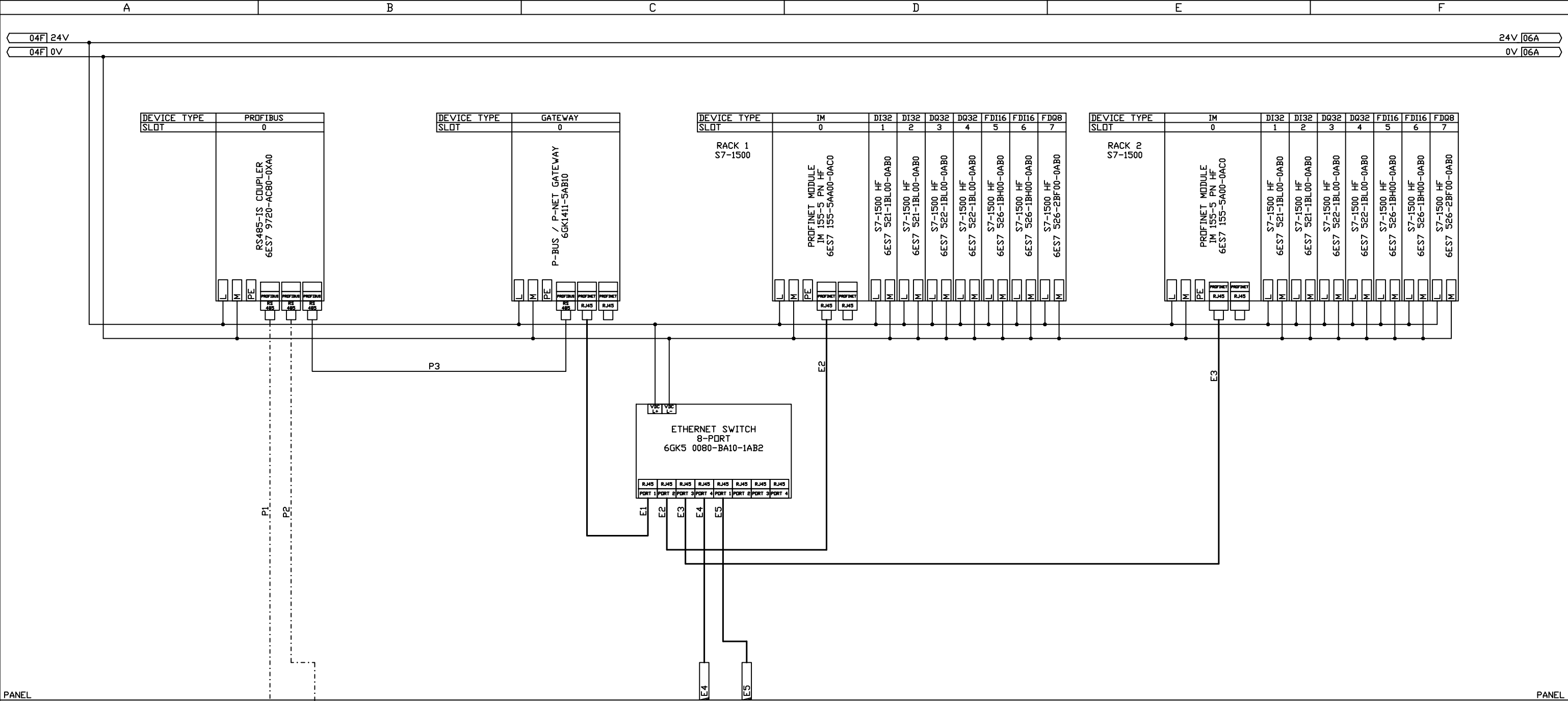
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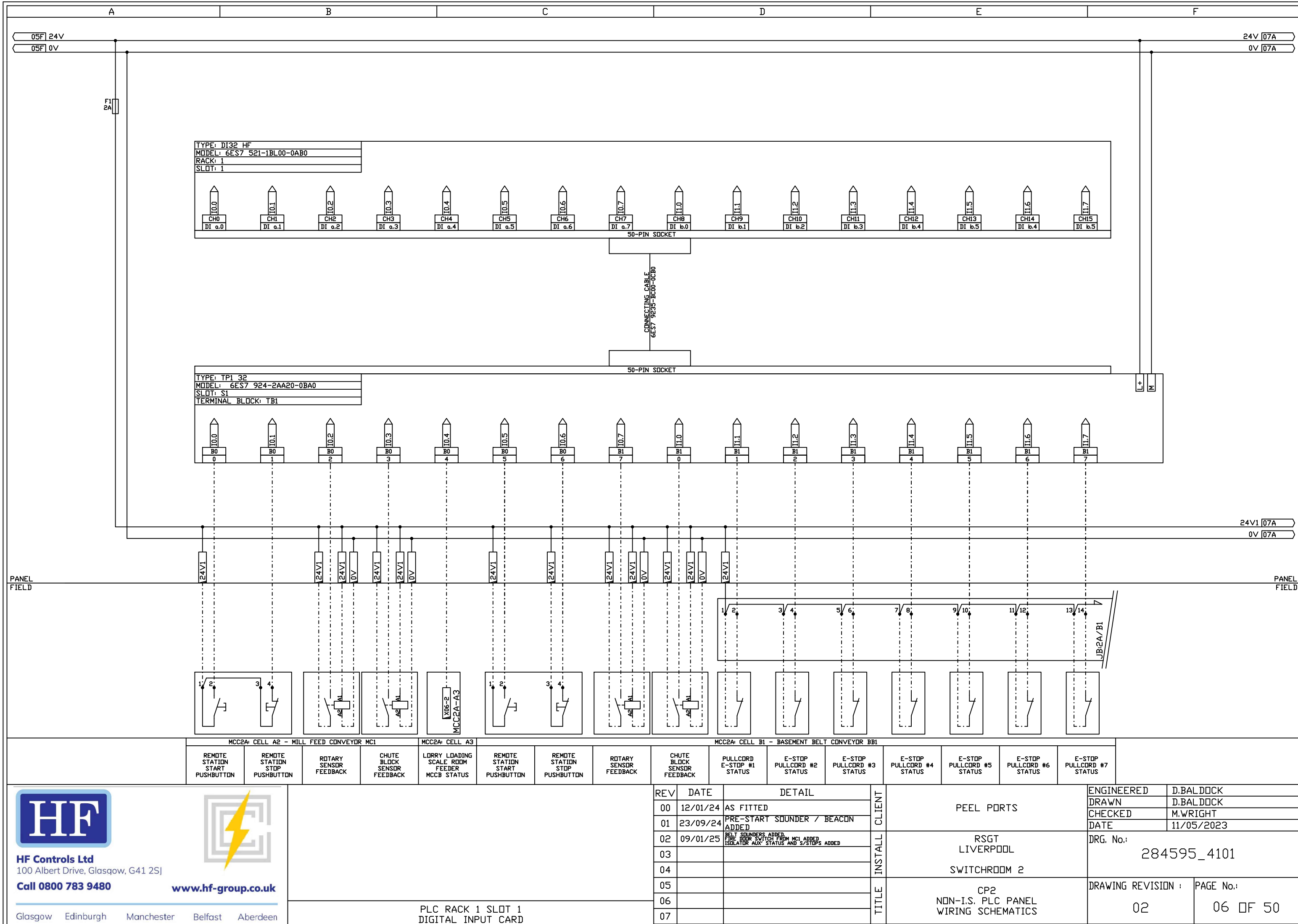
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04			CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	
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06				PAGE No.:	
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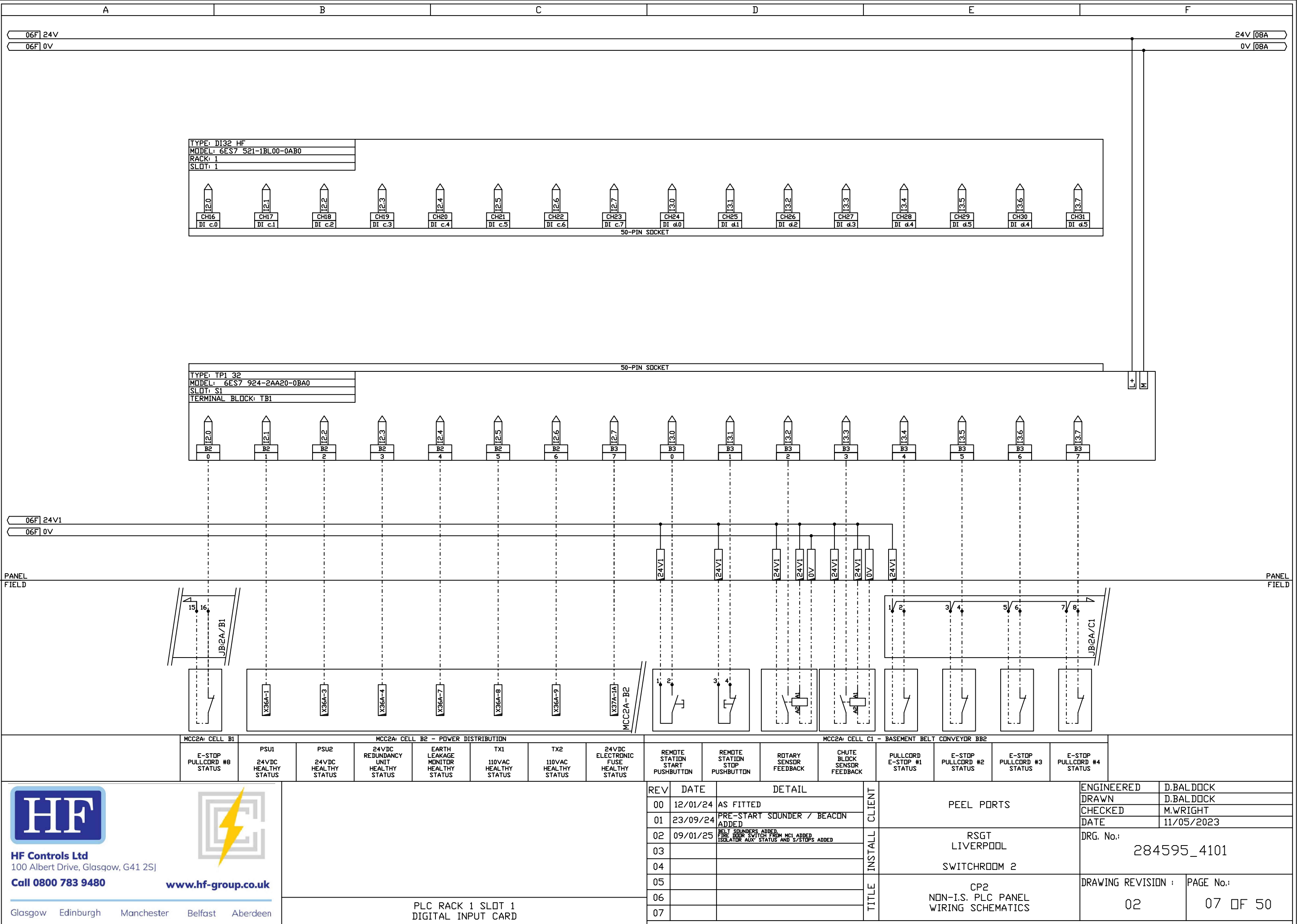


PANEL
FIELD

PANEL
FIELD

		PROFIBUS IN	PROFIBUS OUT			ETHERNET COMMS FROM MCC2A	ETHERNET COMMS FROM MCC2C								
 HF Controls Ltd 100 Albert Drive, Glasgow, G41 2SJ Call 0800 783 9480 www.hf-group.co.uk  Glasgow Edinburgh Manchester Belfast Aberdeen								REV	DATE	DETAIL	CLIENT	PEEL PORTS		ENGINEERED	D.BALDOCK
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								02	09/01/25	BELT SOUNDER ADDED FIRE ALARM SWITCH FROM MCC1 ADDED ISOLATOR AUX STATUS AND S/STOPS ADDED				DATE	11/05/2023
								03			INSTALL	RSGT LIVERPOOL SWITCHROOM 2		DRG. No.: 284595_4101	
								04							
								05							
06			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS		DRAWING REVISION : 02		PAGE No.: 05 OF 50							
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TYPE: TP1 32

MODEL: 6ES7 924-2AA20-0BA0

SLOT: S1

TERMINAL BLOCK: TB1

B2

0

B2

1

B2

2

B2

3

B2

4

B2

5

B2

6

B3

7

B3

0

B3

1

B3

2

B3

3

B3

4

B3

5

B3

6

B3

7

50-PIN SOCKET

06F 24V1

06F 0V

24V1

24V1

24V1

24V1

0V

24V1

24V1

0V

24V1

15

16

JB2A/B1

1

2

3

4

5

6

7

8

JB2A/C1

X36A-1

X36A-3

X36A-4

X36A-7

X36A-8

X36A-9

X37A-1A

MCC2A-B2

1

2

3

4

1

2

1

2

1

2

3

4

5

6

7

8

MCC2A CELL B1

E-STOP PULLCORD #8 STATUS

MCC2A CELL B2 - POWER DISTRIBUTION

PSU1 24VDC HEALTHY STATUS

PSU2 24VDC HEALTHY STATUS

24VDC REDUNDANCY UNIT HEALTHY STATUS

EARTH LEAKAGE MONITOR HEALTHY STATUS

TX1 110VAC HEALTHY STATUS

TX2 110VAC HEALTHY STATUS

24VDC ELECTRONIC FUSE HEALTHY STATUS

MCC2A CELL C1 - BASEMENT BELT CONVEYOR BB2

REMOTE STATION START PUSHBUTTON

REMOTE STATION STOP PUSHBUTTON

ROTARY SENSOR FEEDBACK

CHUTE BLOCK SENSOR FEEDBACK

PULLCORD E-STOP #1 STATUS

E-STOP PULLCORD #2 STATUS

E-STOP PULLCORD #3 STATUS

E-STOP PULLCORD #4 STATUS

HF

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PLC RACK 1 SLOT 1

DIGITAL INPUT CARD

REV

DATE

DETAIL

00

12/01/24

AS FITTED

01

23/09/24

PRE-START SOUNDER / BEACON ADDED

02

09/01/25

BELT SOUNDERS ADDED
FIRE DOOR SWITCH FROM MCC1 ADDED
ISOLATOR AUX STATUS AND S/STOPS ADDED

03

04

05

06

07

CLIENT

PEEL PORTS

ENGINEERED

D.BALDOCK

DRAWN

D.BALDOCK

CHECKED

M.WRIGHT

DATE

11/05/2023

INSTALL

RSGT LIVERPOOL

SWITCHROOM 2

DRG. No.:

284595_4101

TITLE

CP2

NON-I.S. PLC PANEL

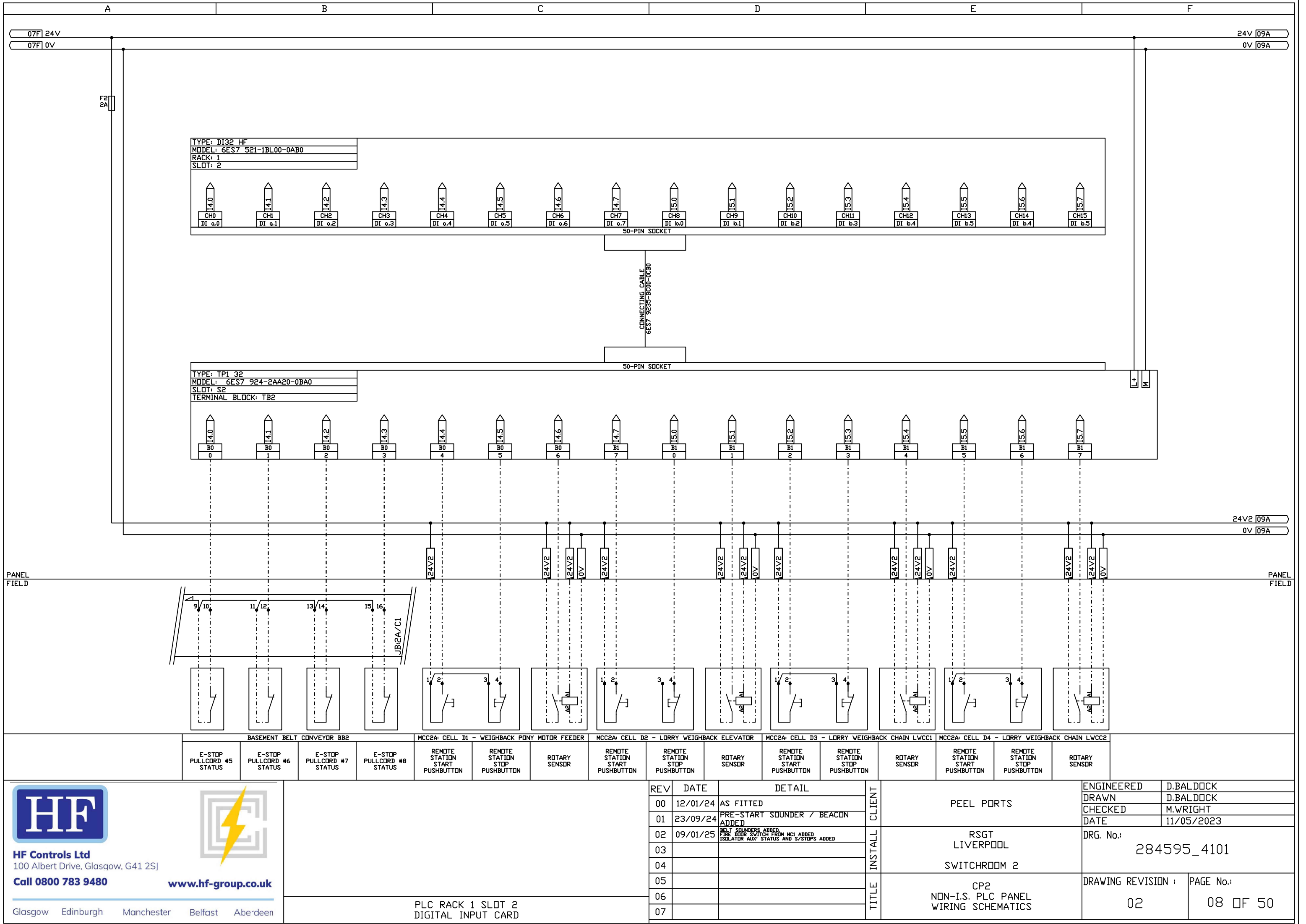
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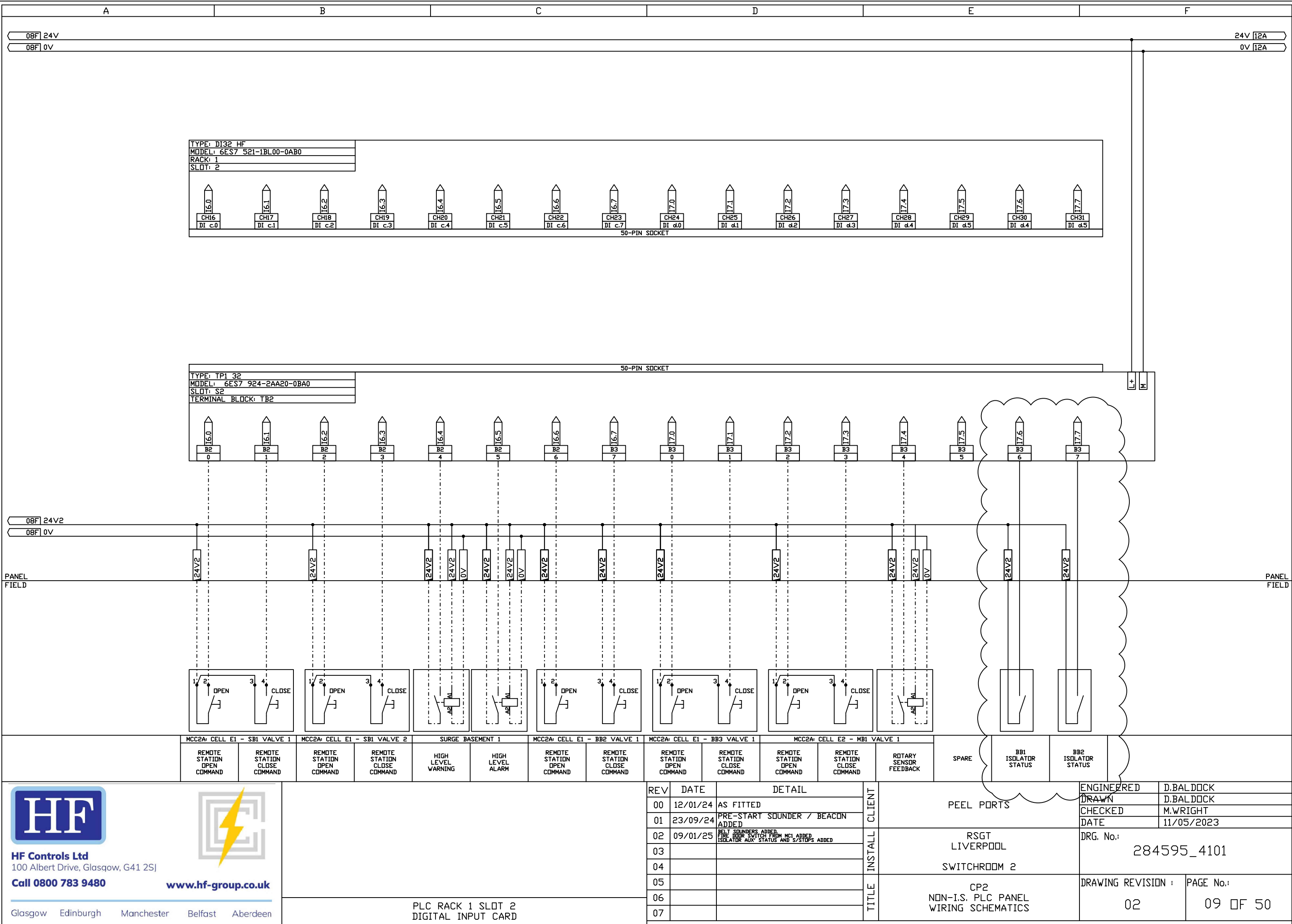
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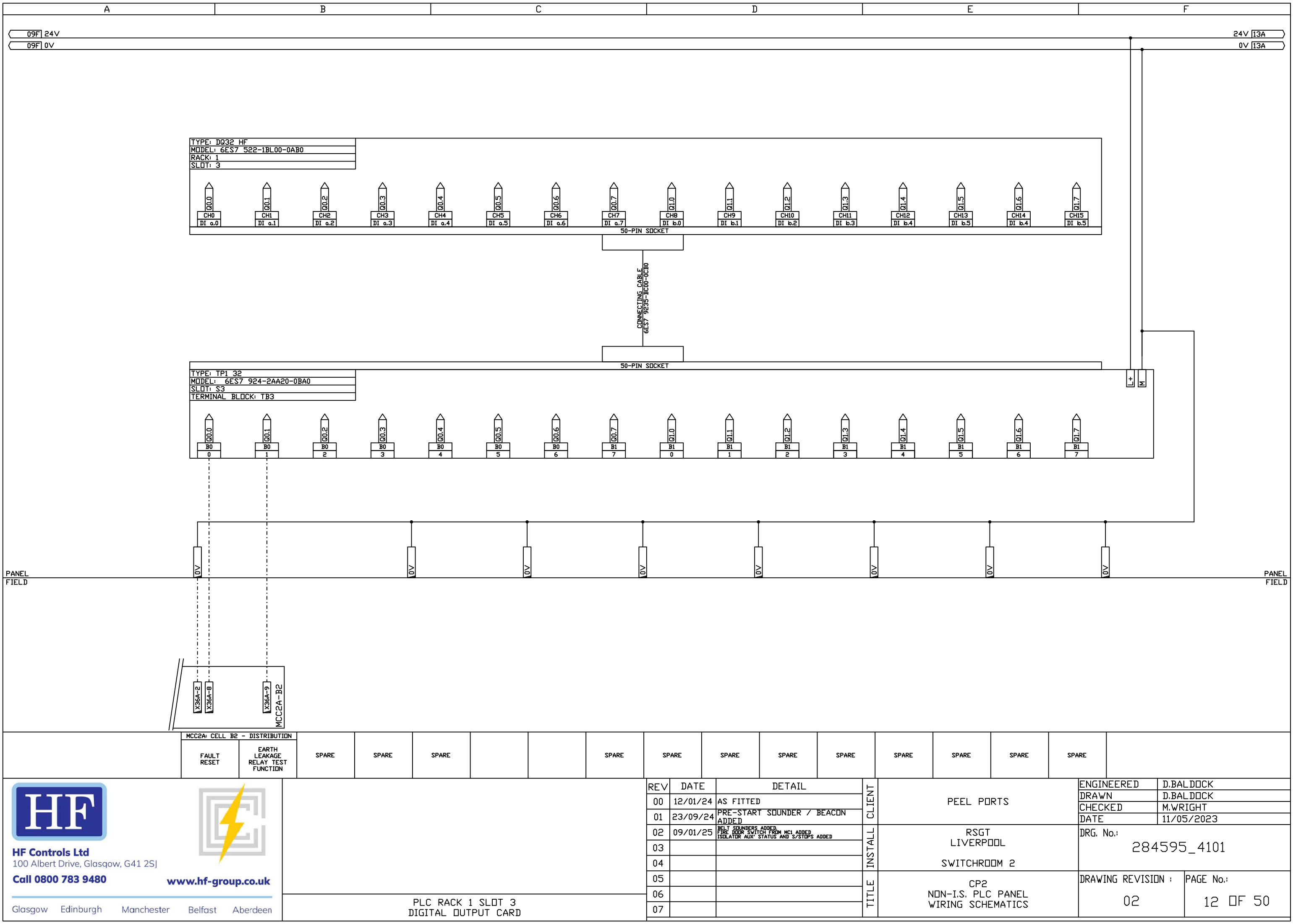
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TYPE: TP1 32

MODEL: 6ES7 924-2AA20-0BA0

SLOT: S3

TERMINAL BLOCK: TB3

Q2.0

B2

0

Q2.1

B2

1

Q2.2

B2

2

Q2.3

B2

3

Q2.4

B2

4

Q2.5

B2

5

Q2.6

B2

6

Q2.7

B3

7

Q3.0

B3

0

Q3.1

B3

1

Q3.2

B3

2

Q3.3

B3

3

Q3.4

B3

4

Q3.5

B3

5

Q3.6

B3

6

Q3.7

B3

7

50-PIN SOCKET

0V

0V

0V

0V

0V

0V

0V

0V

PANEL FIELD

PANEL FIELD

																	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE		
HF HF Controls Ltd 100 Albert Drive, Glasgow, G41 2SJ Call 0800 783 9480 www.hf-group.co.uk Glasgow Edinburgh Manchester Belfast Aberdeen																	REV	DATE	DETAIL		CLIENT	PEEL PORTS			ENGINEERED	D.BALDOCK						
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																	PLC RACK 1 SLOT 3 DIGITAL OUTPUT CARD															

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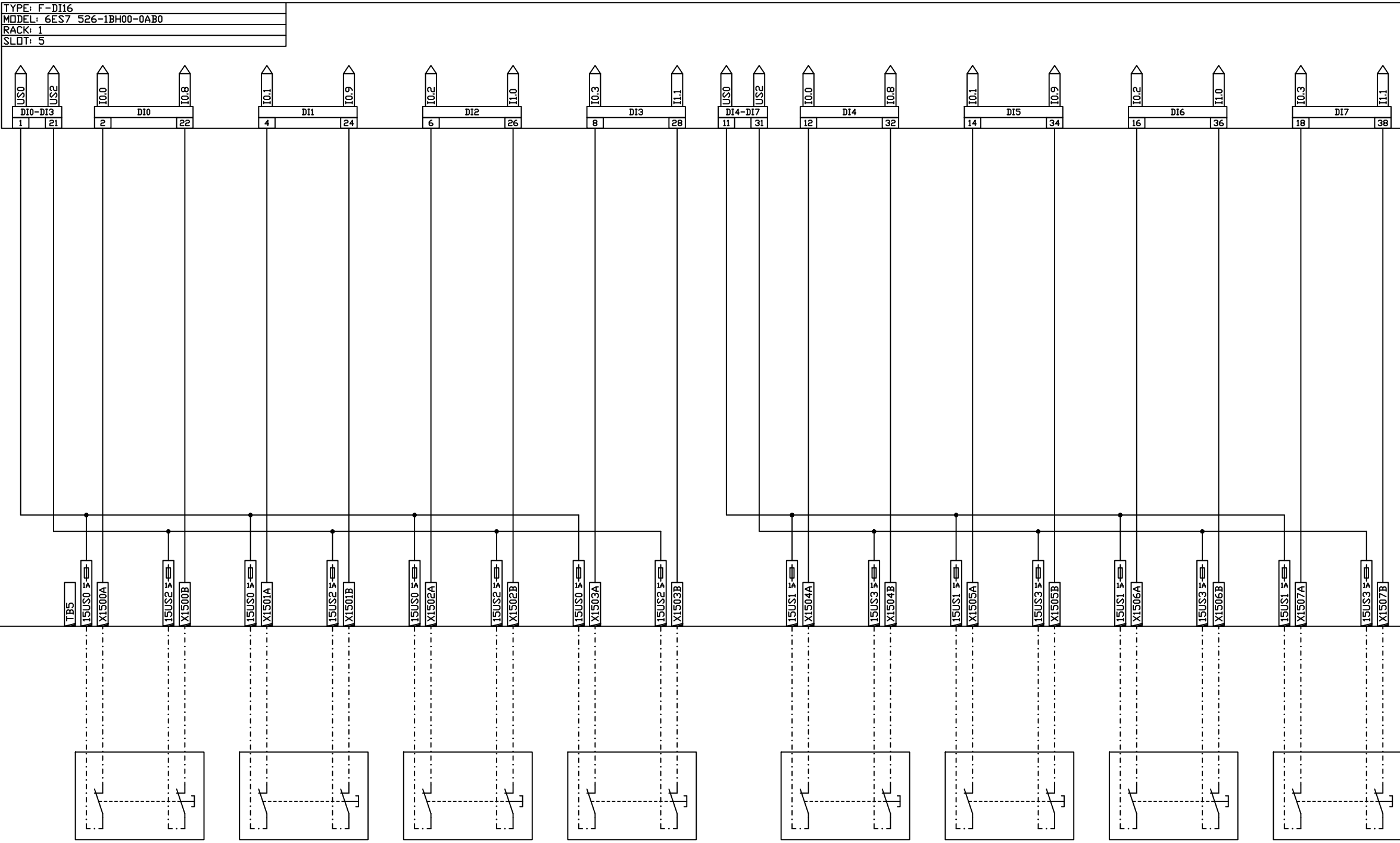
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PANEL
FIELD

PANEL
FIELD

CH0-CH3 SUPPLY		MCC2A: MC1 ESTOP		MCC2A: BB1 ESTOP		MCC2A: BB2 ESTOP		MCC2A: WEIGHBACK PONY ESTOP		CH4-CH7 SUPPLY	MCC2A:WEIGHBACK ELEVTR ESTOP		MCC2A: LWCC1 ESTOP		MCC2A: LWCC2 ESTOP		MCC2A: SBI/V1 ESTOP				
		E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2		E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2					
 HF Controls Ltd 100 Albert Drive, Glasgow, G41 2SJ Call 0800 783 9480 www.hf-group.co.uk Glasgow Edinburgh Manchester Belfast Aberdeen <td colspan="9" rowspan="13"> PLC RACK 1 SLOT 5 SAFETY DIGITAL INPUT CARD </td> <td>REV</td> <td>DATE</td> <td colspan="2">DETAIL</td> <td rowspan="4">CLIENT</td> <td colspan="2" rowspan="4">PEEL PORTS</td> <td colspan="2">ENGINEERED</td> <td colspan="2">D.BALDOCK</td>		PLC RACK 1 SLOT 5 SAFETY DIGITAL INPUT CARD									REV	DATE	DETAIL		CLIENT	PEEL PORTS		ENGINEERED		D.BALDOCK	
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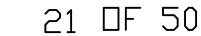
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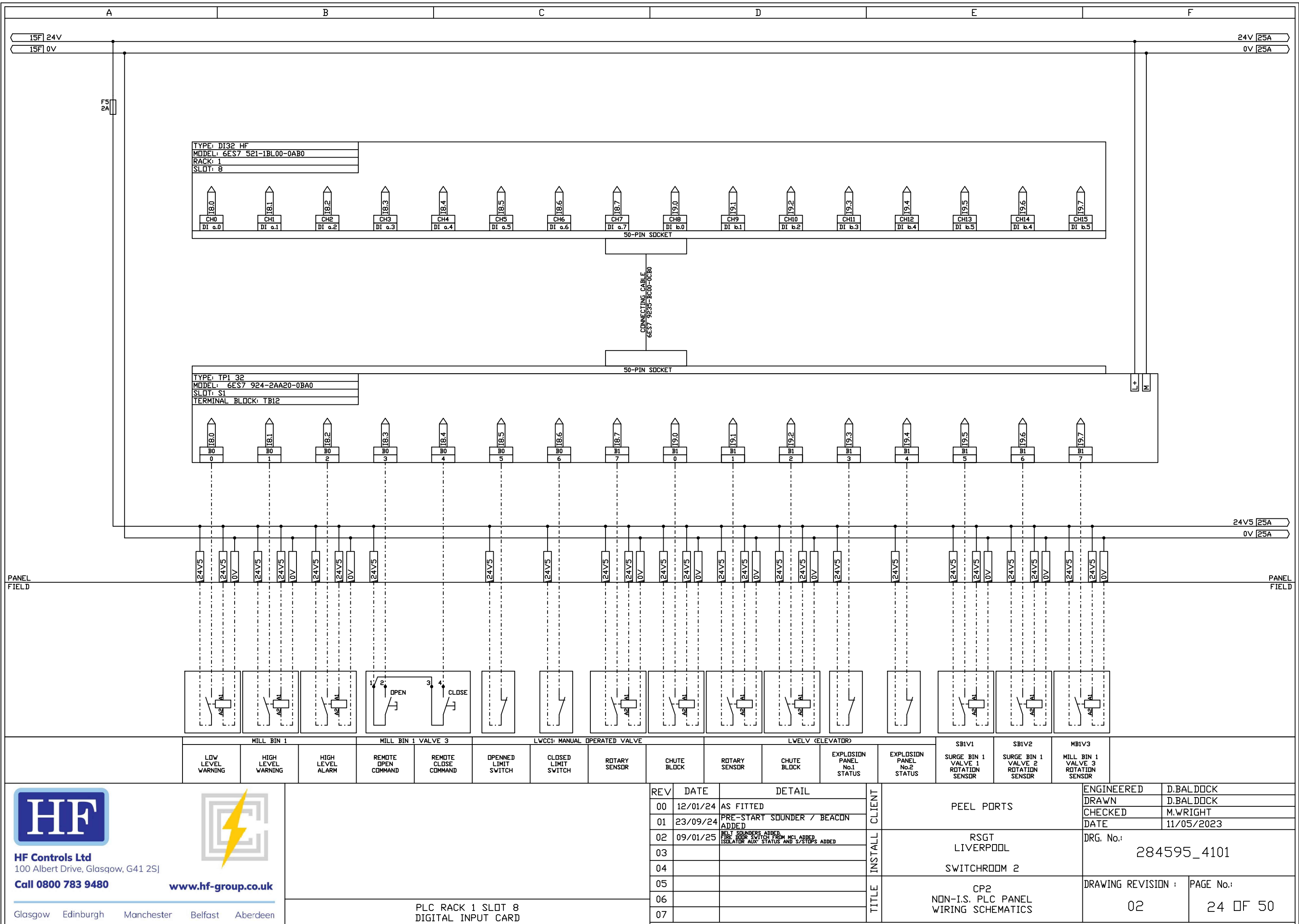
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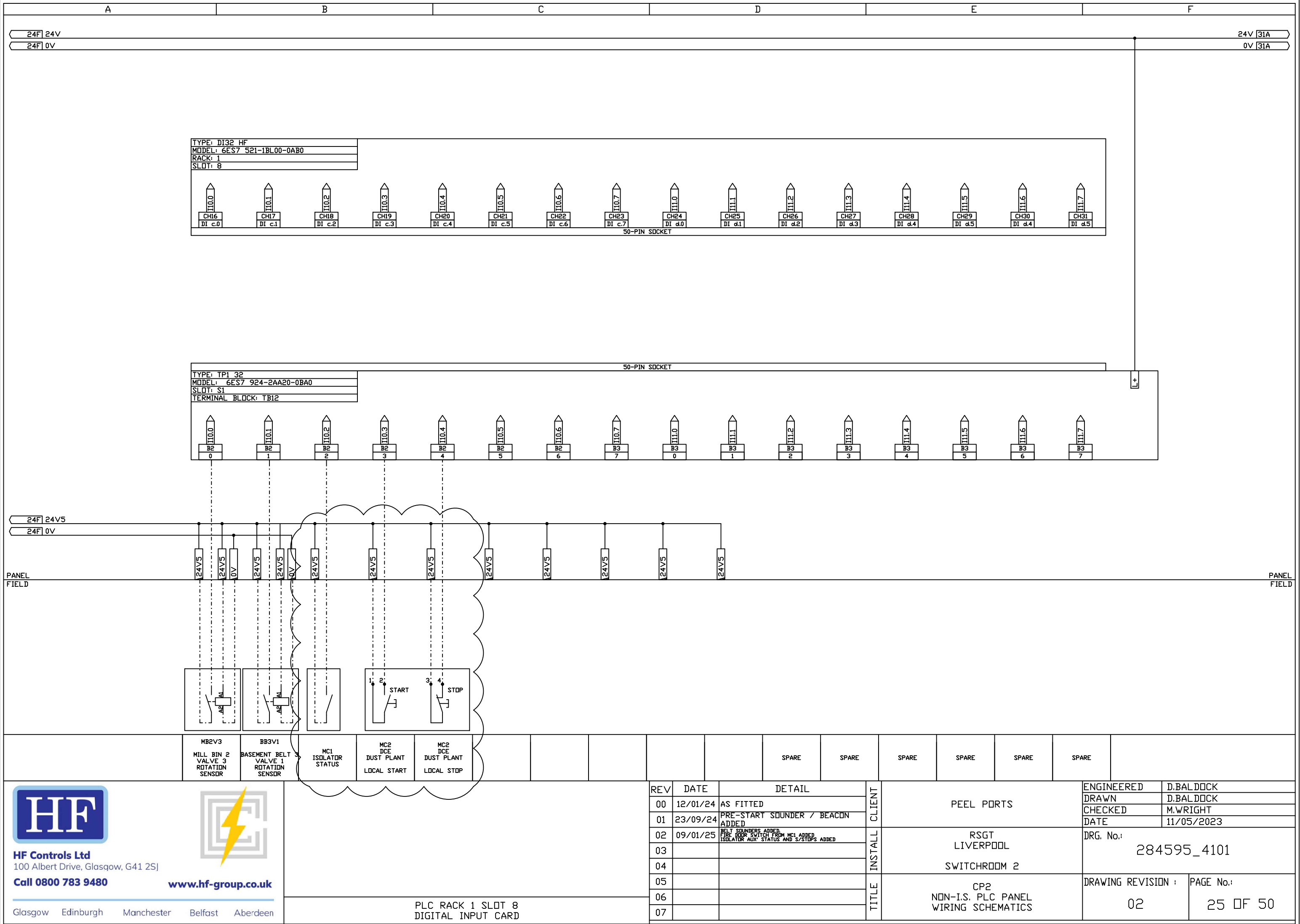
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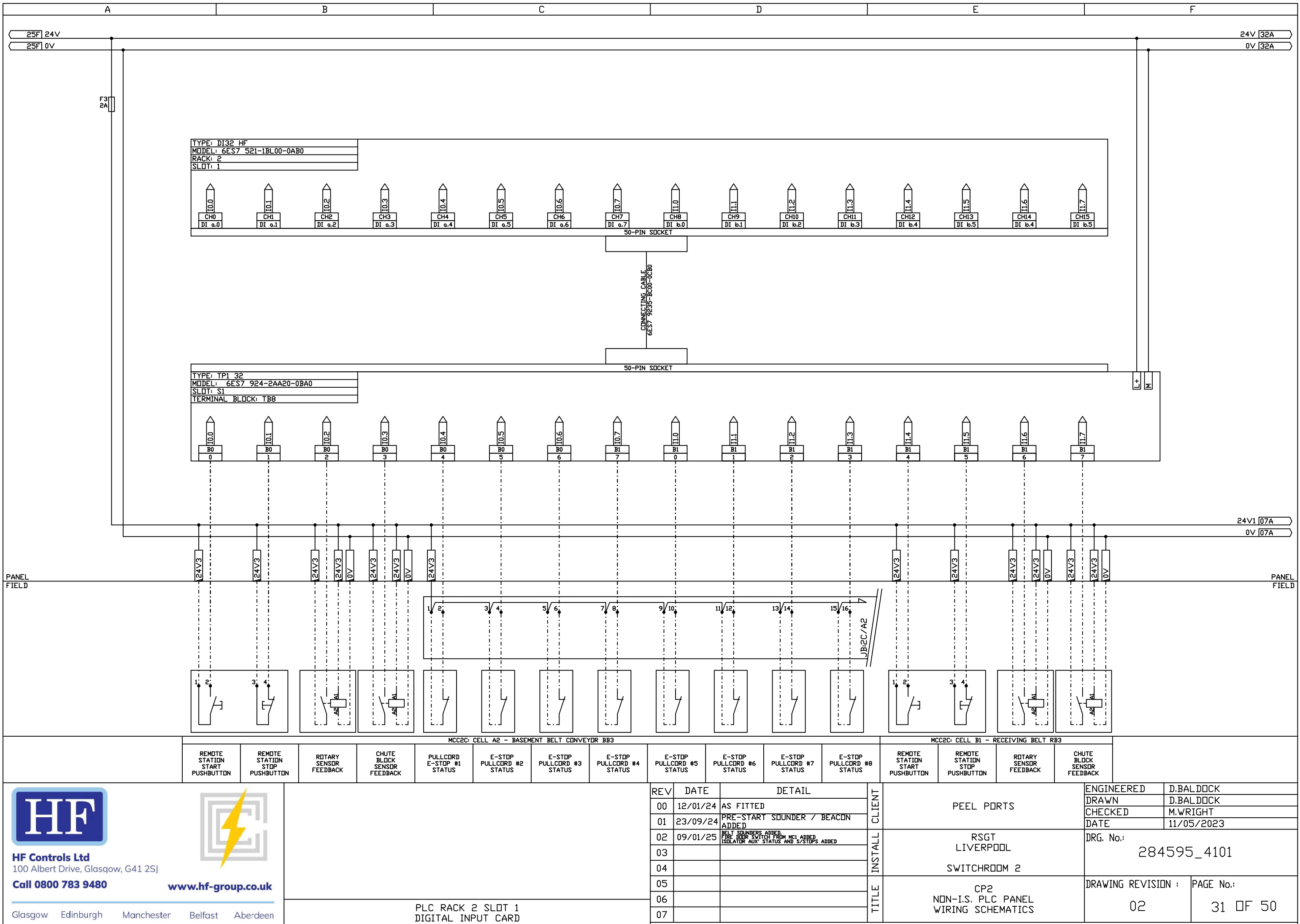
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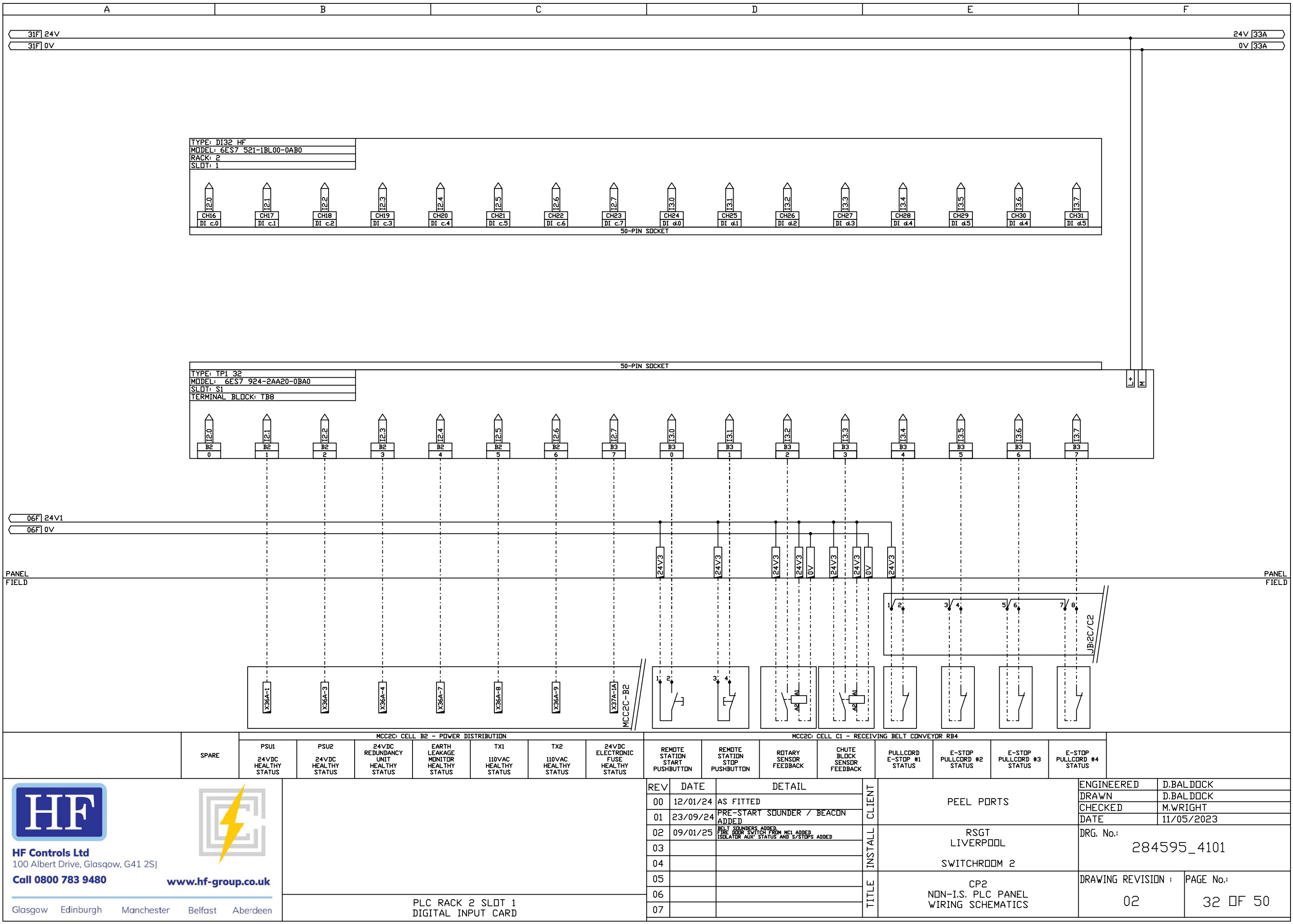


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REV	DATE	DETAIL	CLIENT	ENGINEERED	D.BALDOCK
00	12/01/24	AS FITTED	PEEL PORTS	DRAWN	D.BALDOCK
01	23/09/24	PRE-START SOUNDER / BEACON ADDED		CHECKED	M.WRIGHT
02	09/01/25	BELT SOUNDERS ADDED FIRE DOOR SWITCH FROM MCI ADDED ISOLATOR AUX. STATUS AND S/STOPS ADDED		DATE	11/05/2023
03			RSGT LIVERPOOL SWITCHROOM 2	DRG. No.:	
04				284595_4101	
05					
06			CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	PAGE No.:
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TYPE: TP1 32

MODEL: 6ES7 924-2AA20-0BA0

SLOT: S1

TERMINAL BLOCK: TB8

12.0

B2

0

12.1

B2

1

12.2

B2

2

12.3

B2

3

12.4

B2

4

12.5

B2

5

12.6

B2

6

12.7

B3

7

13.0

B3

0

13.1

B3

1

13.2

B3

2

13.3

B3

3

13.4

B3

4

13.5

B3

5

13.6

B3

6

13.7

B3

7

50-PIN SOCKET

06F 24V1

06F 0V

PANEL FIELD

PANEL FIELD

X36A-1

X36A-3

X36A-4

X36A-7

X36A-8

X36A-9

X37A-1A

MCC2C-B2

1

2

3

4

1

2

1

2

1

2

1

2

1

2

1

2

24V3

24V3

24V3

24V3

0V

24V3

24V3

0V

24V3

24V3

24V3

24V3

24V3

24V3

24V3

24V3

1

2

3

4

1

2

1

2

1

2

1

2

1

2

1

2

JB2C/C2

		MCC2C: CELL B2 - POWER DISTRIBUTION								MCC2C: CELL C1 - RECEIVING BELT CONVEYOR RB4											
	SPARE	PSU1 24VDC HEALTHY STATUS	PSU2 24VDC HEALTHY STATUS	24VDC REDUNDANCY UNIT HEALTHY STATUS	EARTH LEAKAGE MONITOR HEALTHY STATUS	TX1 110VAC HEALTHY STATUS	TX2 110VAC HEALTHY STATUS	24VDC ELECTRONIC FUSE HEALTHY STATUS	REMOTE STATION START PUSHBUTTON	REMOTE STATION STOP PUSHBUTTON	ROTARY SENSOR FEEDBACK	CHUTE BLOCK SENSOR FEEDBACK	PULLCORD E-STOP #1 STATUS	E-STOP PULLCORD #2 STATUS	E-STOP PULLCORD #3 STATUS	E-STOP PULLCORD #4 STATUS					

HF

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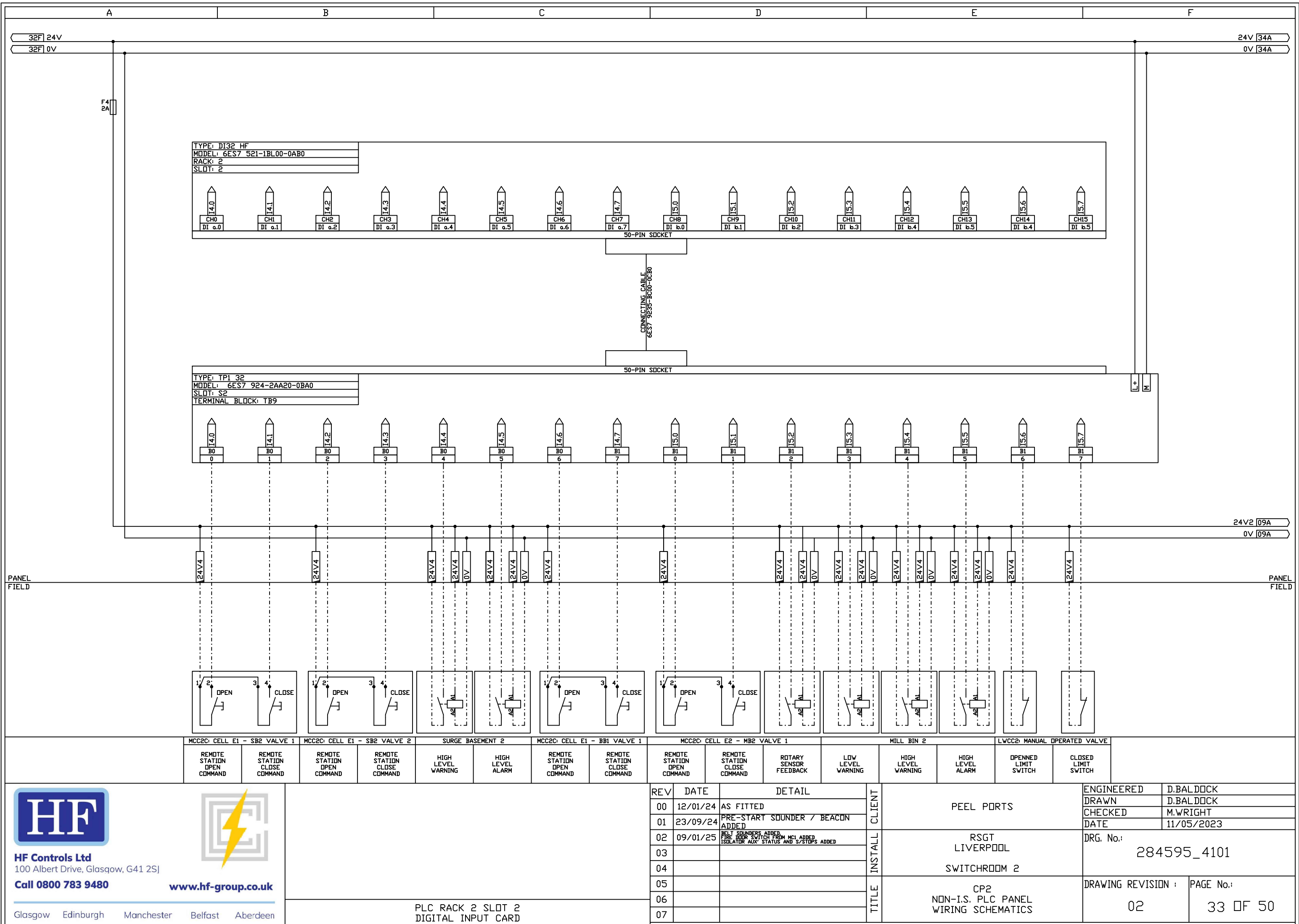
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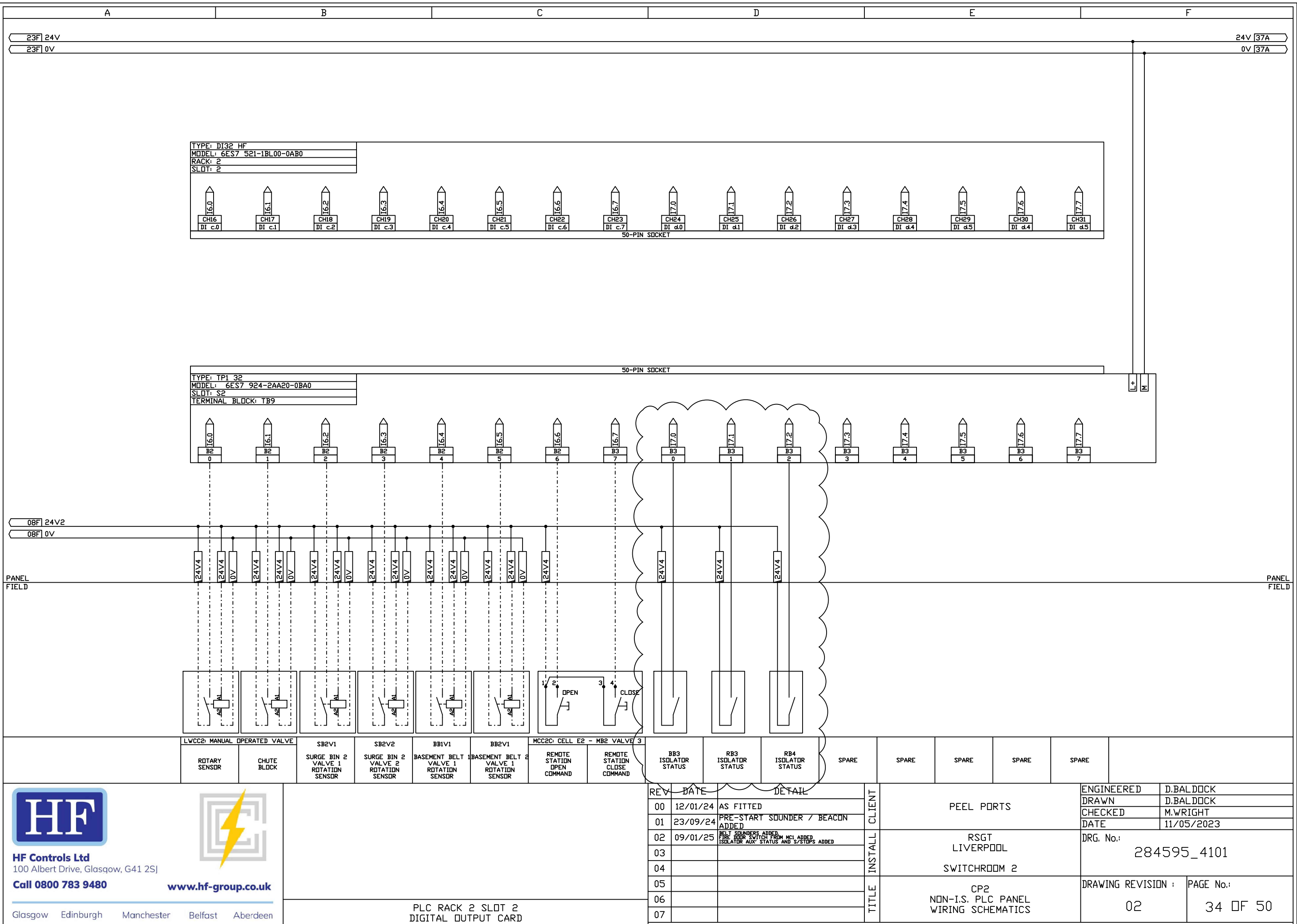
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PLC RACK 2 SLOT 1

DIGITAL INPUT CARD

REV	DATE	DETAIL	CLIENT	INSTALL	TITLE	ENGINEERED D.BAL DOCK	DRAWN D.BAL DOCK	CHECKED M.WRIGHT	DATE 11/05/2023	DRG. No.: 284595_4101	DRAWING REVISION : 02	PAGE No.: 32 OF 50
00	12/01/24	AS FITTED										
01	23/09/24	PRE-START SOUNDER / BEACON ADDED										
02	09/01/25	BELT SOUNDERS ADDED FIRE MOTOR SWITCH FROM MCI ADDED ISOLATOR AUX STATUS AND S/STOPS ADDED										
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REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BALDOCK	
00	12/01/24	AS FITTED			DRAWN	D.BALDOCK	
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02	09/01/25	BELT SOUNDERS ADDED FIRE KICK SWITCH FROM MCI ADDED ISOLATOR AUX STATUS AND S/STOPS ADDED	INSTALL	RSGT LIVERPOOL SWITCHROOM 2	DATE	11/05/2023	
03					DRG. No.:	284595_4101	
04							
05			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	PAGE No.:	
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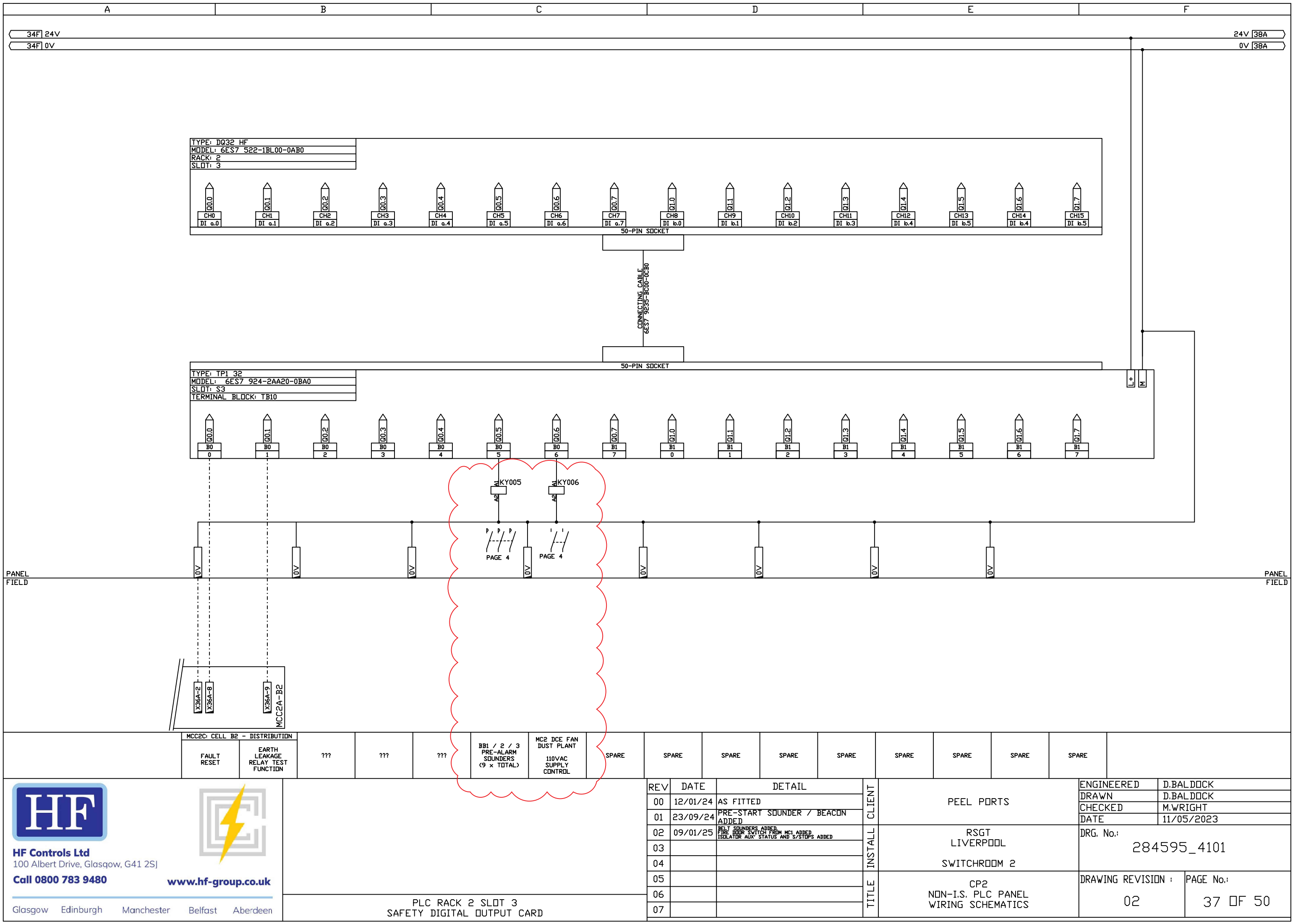
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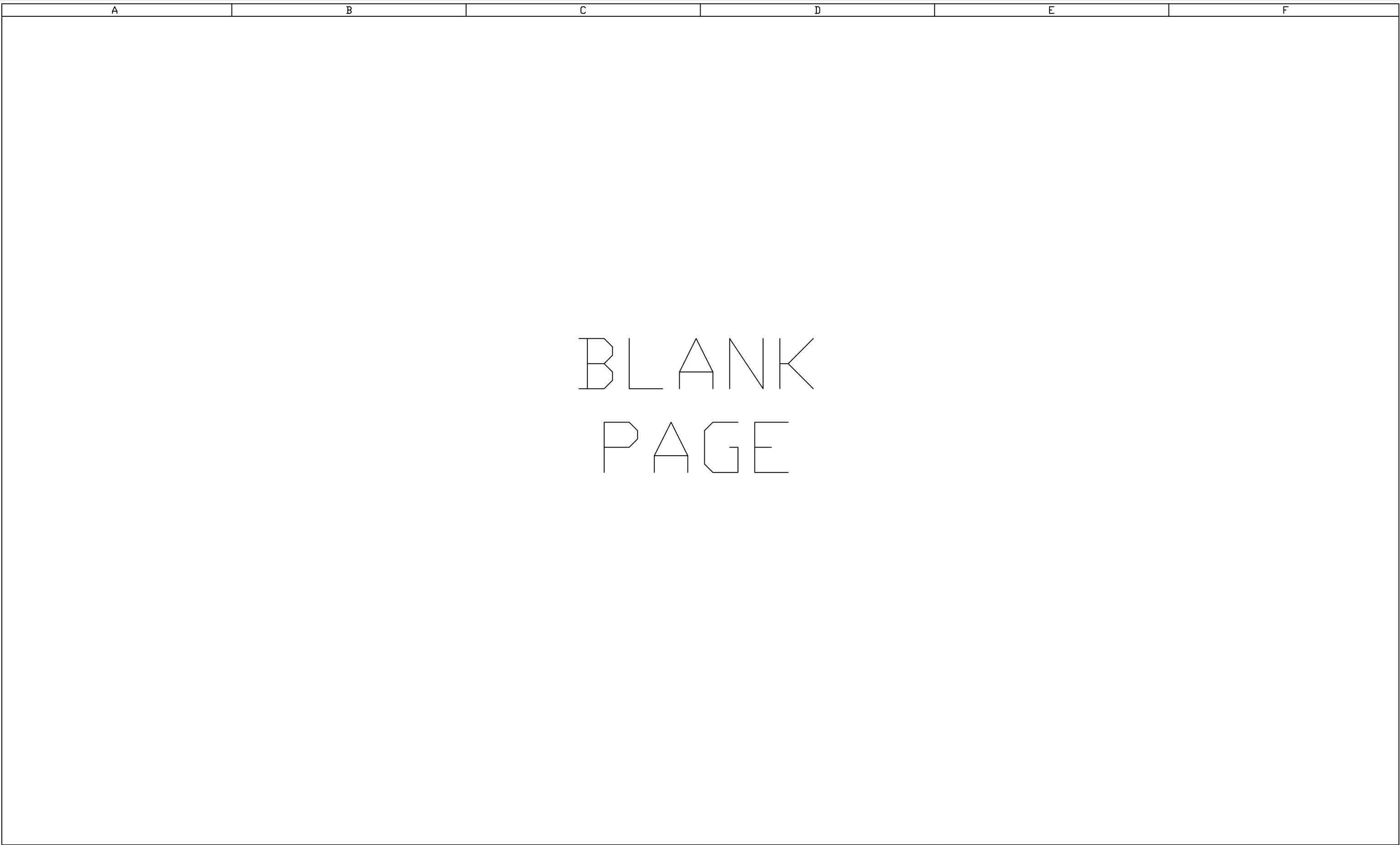
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02	09/01/25	BELT SOUNDERS ADDED FIRE ALARM SWITCH FROM MCL ADDED ISOLATOR AUX STATUS AND S/STOPS ADDED			DATE	11/05/2023
03			INSTALL	RSGT LIVERPOOL SWITCHROOM 2	DRG. No.: 284595_4101	
04						
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06			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	PAGE No.:
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PANEL
FIELD

PANEL
FIELD



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REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BALDOCK	
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02	09/01/25	BELT SOUNDERS ADDED FIRE BEER SWITCH FROM MCI ADDED ISOLATOR AUX STATUS AND S/STOPS ADDED	INSTALL	RSGT LIVERPOOL SWITCHROOM 2	DATE	11/05/2023	
03					DRG. No.: 284595_4101		
04							
05			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	PAGE No.:	
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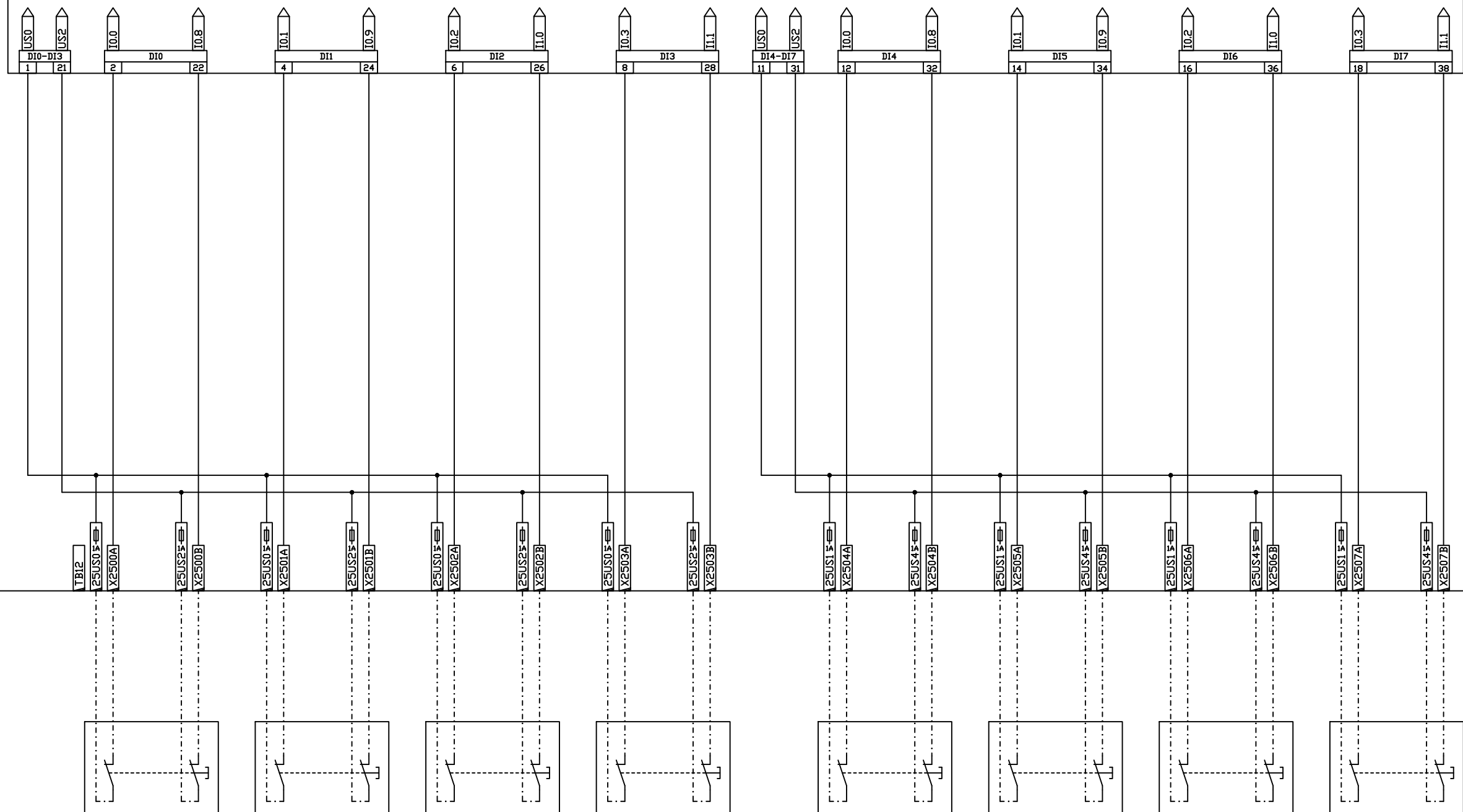
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REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BAL DOCK	
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02	09/01/25	BELT SOUNDERS ADDED FIRE DOOR SWITCH FROM MCI ADDED ISOLATOR AUX STATUS AND S/STOPS ADDED			DATE	11/05/2023	
03			INSTALL	RSGT LIVERPOOL SWITCHROOM 2	DRG. No.: 284595_4101		
04							
05							
06			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :		PAGE No.:
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TYPE: F-DI16
MODEL: 6ES7 526-1BH00-0AB0
RACK: 2
SLOT: 5



CH0-CH3 SUPPLY	MCC2C: BB3 ESTOP		MCC2C: RB3 ESTOP		MCC2C: RB4 ESTOP		MCC2C: SB2-V1		CH4-CH7 SUPPLY	MCC2C: SB2-V2		MCC2C: BB1-V1		MCC2C: MB2-V1		MCC2C: MB2-V3	
	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2		E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2	E-STOP PULLCORD 1 CHANNEL 1	E-STOP PULLCORD 1 CHANNEL 2



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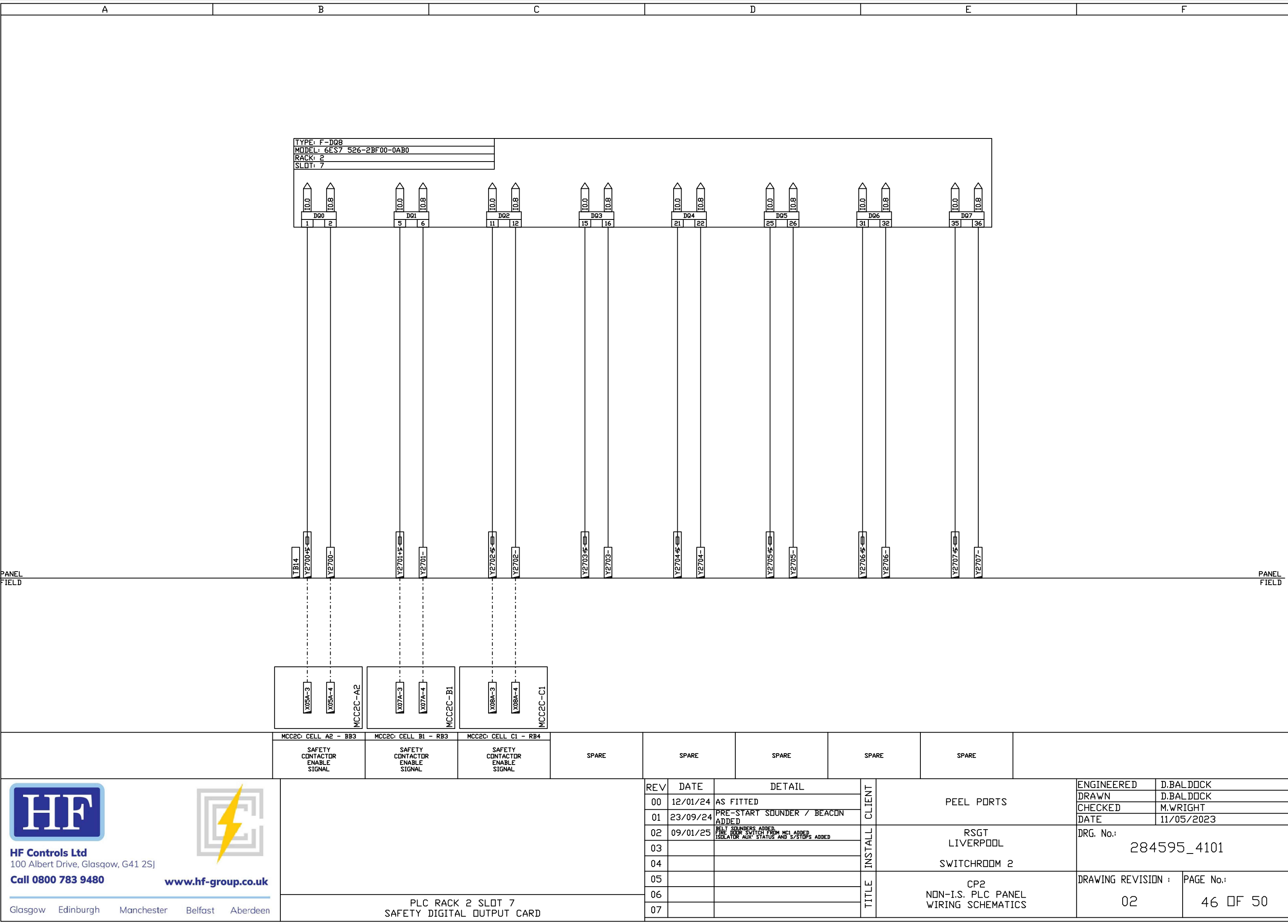
PLC RACK 2 SLOT 5
DIGITAL INPUT CARD

REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BALDOCK
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02	09/01/25	BELT SOUNDERS ADDED FIRE ALARM SWITCH FROM MCL ADDED ISOLATOR AUX STATUS AND S/STOPS ADDED			DATE	11/05/2023
03			INSTALL	RSGT LIVERPOOL SWITCHROOM 2	DRG. No.:	284595_4101
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06			TITLE	NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	PAGE No.:
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REV	DATE	DETAIL	CLIENT	ENGINEERED	D.BALDOCK
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02	09/01/25	BELT SOUNDERS ADDED. FIRE DOOR SWITCH FROM MCI ADDED ISOLATOR AUX STATUS AND S/STEPS ADDED		DATE	11/05/2023
03					
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05				SWITCHROOM 2	
06			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION : 02
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REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BALDOCK
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03			INSTALL	RSGT LIVERPOOL SWITCHROOM 2	DRG. No.: 284595_4101	
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06			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION : 02	PAGE No.: 47 OF 50
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REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BAL DOCK	
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03					DRG. No.:	284595_4101	
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05			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	PAGE No.:	
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REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BAL DOCK	
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03					DRG. No.:		
04					284595_4101		
05			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	PAGE No.:	
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REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BALDOCK	
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03					DRG. No.:	284595_4101	
04							
05			TITLE	CP2 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	PAGE No.:	
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